Cisco Ceiling Microphone Pro Manual

1. Official Cisco Sources

This manual is based on official Cisco documentation for the Cisco Ceiling Microphone Pro, especially the Cisco Ceiling Microphone Pro Data Sheet, the Cisco Microphones Data Sheet, the Cisco Ceiling Microphone Pro Installation Guide, and the Cisco Ceiling Microphone Pro Room Preparation Guidelines.

The source manual identifies the Cisco Ceiling Microphone Pro as an intelligent IP ceiling microphone with AI adaptive beamforming, designed for Cisco collaboration systems and managed through the connected Cisco video device and Cisco Control Hub.

Practical installation, placement, verification, troubleshooting, and maintenance notes were added to make this manual more useful for real deployment and support work. Exact behavior may vary depending on the connected Cisco video device, RoomOS version, room layout, mounting option, cable quality, and Control Hub deployment.

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2. Device Description

The Cisco Ceiling Microphone Pro is a digital IP ceiling microphone designed for Cisco meeting room and collaboration systems.

It captures speech from participants in the room using intelligent adaptive beamforming. The microphone focuses on active speakers while helping suppress background noise and unwanted room sounds.

It is commonly used in:

- Meeting rooms
- Boardrooms
- Training rooms
- Classrooms
- Multi-purpose rooms
- Collaboration spaces
- Hybrid meeting environments

In a Cisco collaboration room, the Ceiling Microphone Pro should be treated as the **room audio-capture peripheral**. It connects to a Cisco video device using a single Ethernet/IP cable and works with the video device to provide high-quality voice pickup.

The microphone is not a standalone recording device, speaker, camera, network switch, or general audio DSP. Its role is voice capture for conferencing and collaboration.

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3. Technical Specifications

The Cisco Ceiling Microphone Pro is manufactured by Cisco and is designed as an intelligent IP ceiling microphone.

Main characteristics include:

- Digital IP conferencing microphone
- Ceiling-mounted design
- AI adaptive beamforming
- Eight adaptive beams
- Embedded signal processing
- No external DSP required
- Single Ethernet/IP cable connection
- Power and audio over the same connection
- Managed through Cisco video device
- Managed and monitored through Cisco Control Hub
- Zero-touch configuration
- Secure encryption
- Uses visual data from Cisco cameras to optimize pickup
- Supports camera tracking workflows when used in compatible Cisco systems
- Pickup radius up to 3.5 m / 11.5 ft
- One-year limited warranty

Power and connection:

```
Connection type: Single Ethernet/IP cable to Cisco video device
Power: Delivered through the single-cable connection
Safety requirement: Connect only to LPS/PS2-compliant I/O
Cable: Sold separately
```

Wireless support:

```
Wireless support: Not supported
```

Mounting options:

```
Drop-ceiling grid mount
Ceiling bracket mount
Wire-hanging mount
Mounting kits are sold separately
```

Operating environment and physical dimensions:

```
Operating environment: Not found in the provided official-source manual
Physical dimensions: Not found in the provided official-source manual
```

Important note:

The microphone should be handled carefully because the source manual identifies it as fragile during installation and handling.

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4. Ports, Mounting, and Indicators

The Ceiling Microphone Pro has a simple physical connection design but depends heavily on correct placement and installation.

4.1 Single Ethernet/IP Connection

The microphone uses one Ethernet/IP cable connection to the Cisco video device.

This cable carries:

- Power
- Audio data
- Device communication
- Management/control communication

Important safety note:

The microphone must only be connected to LPS/PS2-compliant I/O.

4.2 Beamforming Microphone System

Internally, the device uses eight adaptive beams.

The beamforming system is used to:

- Focus on active speakers
- Reduce background noise
- Improve speech clarity
- Capture voices across the room
- Support intelligent room audio processing

4.3 Status Indicator LED

The microphone includes a status indicator LED.

The LED can help identify:

- Power status
- Connection status
- Microphone status
- Error or abnormal state

During troubleshooting, check the LED status before replacing hardware or changing room configuration.

4.4 Mounting Hardware

The microphone can be installed using supported mounting options:

- Drop-ceiling grid mount
- Ceiling bracket mount
- Wire-hanging mount

Mounting hardware is sold separately and should match the room ceiling type and installation design.

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5. Room Planning and Placement

Placement is critical for good microphone performance.

Before installation, review room size, ceiling height, table position, participant seating, HVAC noise, projector noise, speaker placement, glass walls, reflective surfaces, and camera position.

The microphone supports voice pickup up to a radius of approximately:

```
3.5 m / 11.5 ft
```

Placement should aim to cover active speaking positions, not empty areas.

Recommended placement considerations:

- Install above or near the main meeting table.
- Keep within the documented pickup radius.
- Avoid placing too close to HVAC vents.
- Avoid placing directly above loud equipment.
- Avoid placing near ceiling speakers if echo/noise is a concern.
- Avoid blocked locations.
- Use room preparation guidelines for coverage planning.
- Align installation with Cisco camera and room intelligence design where applicable.

Poor placement can cause:

- Weak pickup
- Uneven voice capture
- Excessive background noise
- Poor far-end audio experience
- Unstable speaker tracking
- Reduced conferencing quality

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6. Installation and Setup Notes

Before installation, confirm the microphone model, Cisco video device compatibility, mounting hardware, cable path, room layout, pickup coverage, and safety requirements.

Basic installation steps:

1. Review the Cisco Room Preparation Guidelines.

2. Select the microphone mounting location.
3. Confirm the location covers the intended participant area.
4. Select the appropriate mount: drop-ceiling, bracket, or wire-hanging.
5. Handle the microphone carefully.
6. Install the mount securely according to the installation guide.
7. Connect the Ethernet/IP cable to the microphone.
8. Route the cable safely to the Cisco video device.
9. Connect only to LPS/PS2-compliant I/O.
10. Power the system through the connected video device.
11. Verify that the microphone powers on.
12. Verify that the connected video device detects the microphone.
13. Check microphone status through the video device or Control Hub.
14. Run test calls or audio tests.
15. Confirm clear pickup from all expected seating positions.
16. Document microphone location, cable path, connected port, and room coverage.

Important installation notes:

- The Ethernet/IP cable is sold separately.
- Mounting kits are sold separately.
- The microphone is fragile and should be handled carefully.
- Incorrect placement can cause poor audio even if the hardware is working.
- The microphone should be managed as part of the Cisco video device system.

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7. Connection to Cisco Video Device

The Ceiling Microphone Pro is designed to work with a Cisco video device.

The connected video device provides:

- Power
- Audio integration
- Device management
- Software updates
- Control Hub reporting
- Room intelligence integration
- Camera-assisted optimization where supported

The microphone should appear as a connected peripheral or audio device through the Cisco video endpoint and management tools.

If the microphone does not appear, check cable, port, compatibility, RoomOS version, and LPS/PS2 compliance.

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8. Normal Operation

During normal operation:

- The microphone receives power through the IP connection.
- Status indicator shows normal state.
- Cisco video device detects the microphone.
- Microphone appears in device diagnostics or Control Hub where available.
- Eight adaptive beams capture room speech.
- Active speakers are picked up clearly.
- Background noise is reduced.
- Audio is transmitted to the far end during calls.
- Cisco camera visual data may help optimize pickup.
- Camera tracking and room intelligence features work where supported.
- No repeated connection or microphone errors appear in diagnostics.

Useful normal-operation checks:

```
Check microphone status LED
Check cable connection
Check connected Cisco video device peripheral status
Check Control Hub status if used
Place a test call
Check far-end audio quality
Walk around expected seating positions and test pickup
Check for background noise or echo
Verify camera tracking behavior if used
```

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9. Troubleshooting

Troubleshooting should move from physical installation to system-level checks:

1. Cable and connection
2. LPS/PS2-compliant I/O
3. Video device compatibility
4. RoomOS/software version
5. Peripheral detection
6. Placement and pickup radius
7. Room noise and acoustics
8. Camera integration
9. Control Hub status
10. Hardware fault

9.1 Microphone Not Detected

Symptom:

- Microphone does not appear on the Cisco video device
- Control Hub does not show the microphone
- No audio input from microphone

- Status LED abnormal or off

Possible causes:

- Cable disconnected
- Bad Ethernet/IP cable
- Connected to wrong port
- Non-compliant I/O connection
- Video device does not support the microphone
- Software version issue
- Microphone hardware fault

Checks:

```
Check Ethernet/IP cable
Reseat both ends of cable
Verify connection to Cisco video device
Confirm LPS/PS2-compliant I/O
Check video device peripheral status
Check Control Hub
Check RoomOS version
Try known-good cable if available
```

Fix actions:

- Reseat or replace the Ethernet/IP cable.
- Connect to the correct Cisco video device port.
- Use only compliant I/O.
- Update connected video device software if required.
- Confirm device compatibility.
- Escalate if microphone remains undetected with known-good cable and supported device.

9.2 No Audio Pickup

Symptom:

- Far-end cannot hear room
- Microphone appears connected but no speech is transmitted
- Audio level is missing
- Test call has no microphone audio

Possible causes:

- Microphone muted in system
- Wrong audio input selected
- Video device configuration issue
- Cable issue
- Microphone not fully detected

- Call audio settings incorrect
- Hardware fault

Checks:

```
Check mute state
Check video device audio input settings
Check microphone detected status
Check call audio path
Test with another call
Check cable connection
Check logs/diagnostics
```

Fix actions:

- Unmute system.
- Select correct microphone/input configuration.
- Reseat cable.
- Restart connected video device if required.
- Update device software.
- Escalate if no audio is detected after configuration and cabling are confirmed.

9.3 Weak or Uneven Voice Pickup

Symptom:

- Some participants sound quiet
- Far-end reports weak audio
- Pickup is good only near one area
- Audio quality changes depending on seat position

Possible causes:

- Microphone placed too far from participants
- Seating outside 3.5 m / 11.5 ft pickup radius
- Incorrect ceiling location
- Room layout changed after installation
- Ceiling height or acoustics unsuitable
- Background noise masks speech
- Mounting location not aligned with speaking area

Checks:

```
Check participant positions
Measure distance to microphone
Compare with 3.5 m / 11.5 ft pickup radius
Check room layout
```

```
Check ceiling placement
Test speech from each seat
```

Fix actions:

- Reposition microphone within proper coverage.
- Add additional microphones if room coverage requires it.
- Follow Room Preparation Guidelines.
- Reduce background noise.
- Adjust room layout if practical.
- Confirm camera-assisted optimization is working where supported.

9.4 Background Noise or Poor Audio Quality

Symptom:

- Far-end hears HVAC noise
- Audio sounds noisy or unclear
- Echo or room noise affects call quality
- Speech is hard to understand

Possible causes:

- Microphone installed near HVAC vent
- Loud equipment nearby
- Poor room acoustics
- Hard reflective surfaces
- Ceiling speakers too close
- Room noise too high
- Microphone placement issue

Checks:

```
Identify noise sources
Check HVAC vents
Check projector/fan noise
Check speaker placement
Check room acoustics
Run test call
Compare audio with room quiet and noisy
```

Fix actions:

- Move microphone away from noise sources.
- Reduce HVAC noise if possible.
- Improve room acoustics.
- Adjust speaker placement.

- Use Cisco room preparation guidance.
- Add acoustic treatment if needed.

9.5 Peripheral Not Managed in Control Hub

Symptom:

- Microphone works locally but does not appear in Control Hub
- Peripheral status missing
- Remote monitoring unavailable

Possible causes:

- Video device not registered or not cloud-managed
- Control Hub access issue
- Microphone not associated correctly
- Software version issue
- Network/service connectivity issue

Checks:

```
Check video device Control Hub status
Check device registration
Check peripheral list
Check network reachability
Check software version
```

Fix actions:

- Verify video device is registered and online.
- Confirm microphone is detected by video device.
- Update video device software.
- Check Control Hub permissions and device assignment.

9.6 Camera Tracking or Speaker Optimization Not Working

Symptom:

- Camera does not track active speaker correctly
- Audio pickup does not seem optimized
- Speaker tracking is inconsistent

Possible causes:

- Cisco camera visual data unavailable
- Camera not connected or not configured
- Microphone placement not aligned with camera view
- Room layout changed

- Software feature not enabled or supported
- Poor lighting or visual obstruction

Checks:

```
Check Cisco camera status
Check camera view
Check microphone placement
Check RoomOS configuration
Check room layout
Check Control Hub diagnostics
```

Fix actions:

- Verify camera is connected and working.
- Align microphone placement with room coverage.
- Adjust camera view.
- Update software.
- Confirm feature support on the connected video device.

9.7 Connection or Safety Issue

Symptom:

- Microphone unstable
- Device not detected
- Safety or power concern
- Unexpected behavior after connection

Possible cause:

- Connected to non-LPS/PS2-compliant I/O

Fix:

```
Disconnect from non-compliant connection
Connect only to LPS/PS2-compliant I/O
Verify operation after reconnecting correctly
```

Important warning:

Do not bypass the safety requirement. Use only supported and compliant connections.

9.8 Physical Damage During Installation

Symptom:

- Microphone housing damaged
- Mounting points damaged
- Microphone fails after handling
- Device appears loose or unstable

Possible causes:

- Mishandling
- Dropping the microphone
- Incorrect mount
- Excessive force during installation
- Unsupported installation method

Fix actions:

- Stop installation if damage is visible.
- Inspect mounting hardware.
- Use only supported mounting options.
- Replace damaged unit if required.
- Handle future units carefully.

9.9 Software or Firmware-Related Issue

Symptom:

- Microphone detected but behaves unexpectedly
- Known bug suspected
- Peripheral update needed
- Control Hub shows issue

Possible causes:

- Connected Cisco video device software is outdated
- Peripheral firmware update pending
- RoomOS compatibility issue
- Configuration issue

Fix actions:

```
Check connected Cisco video device software version
Update RoomOS or device software
Allow peripheral updates through the video device
Verify microphone after update
Check Control Hub diagnostics
```

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10. Factory Reset and Recovery

Factory reset and recovery steps for the Cisco Ceiling Microphone Pro were not found in the provided official-source manual.

The microphone is managed and updated through the connected Cisco video device and Cisco Control Hub.

Recommended recovery approach:

1. Check cable and connection first.
2. Verify LPS/PS2-compliant I/O.
3. Verify the microphone is detected by the Cisco video device.
4. Check Control Hub status if deployed.
5. Restart or power-cycle through the connected video device if required.
6. Update the connected video device software.
7. Replace cable if detection remains unstable.
8. Escalate to Cisco support if the microphone remains undetected with known-good cable and supported device.

Important note:

Do not invent or attempt undocumented reset procedures unless confirmed by official Cisco documentation for the exact product and software version.

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11. Maintenance and Best Practices

Recommended practices:

- Keep the connected Cisco video device software current.
- Check microphone status through the video device and Control Hub.
- Use the Room Preparation Guidelines before installation.
- Place the microphone within the correct pickup radius.
- Avoid placing microphone near HVAC vents or noisy equipment.
- Use supported mounting hardware.
- Handle the microphone carefully.
- Keep the microphone clean and dust-free.
- Do not use harsh cleaning chemicals.
- Check mounting security periodically.
- Inspect the Ethernet/IP cable.
- Label the cable and connected video device port.
- Document microphone location and coverage area.
- Test audio quality after room layout changes.
- Re-test audio after software updates.
- Power-cycle only through the connected video system when required.

Useful maintenance checks:

```
Check microphone detection
Check audio pickup from all seating positions
Check Control Hub status
```

```
Check cable condition
Check mounting hardware
Check room noise
Check Cisco video device software version
Check camera tracking behavior if used
```

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12. Quick Reference

Power Check

```
Verify single Ethernet/IP cable connection
Confirm connected Cisco video device is powered
Confirm connection is LPS/PS2-compliant
Check microphone status LED
```

Detection Check

```
Check Cisco video device peripheral status
Check Control Hub if deployed
Reseat Ethernet/IP cable
Try known-good cable
Update video device software if needed
```

Audio Pickup Check

```
Test call audio
Check mute state
Speak from each seat
Confirm seating is within 3.5 m / 11.5 ft radius
Check room noise
```

Placement Check

```
Follow Room Preparation Guidelines
Avoid HVAC vents
Avoid loud equipment
Place above intended speaking area
Confirm coverage radius
```

Cable Check

```
Inspect Ethernet/IP cable
Check both ends
Replace cable if damaged
Use correct route and port
```

Control Hub Check

```
Confirm video device is online
Confirm microphone appears as peripheral
Check diagnostics and alerts
```

Camera Optimization Check

```
Check Cisco camera status
Verify camera visual data is available
Check room layout and microphone placement
Confirm feature support
```

Restart / Recovery

```
Power-cycle through connected Cisco video device if required
Do not use undocumented reset steps
Update connected video device software
Escalate if still undetected
```

Factory Reset

```
Factory reset steps were not found in the provided official-source manual
Manage and update through Cisco video device and Control Hub
```
